

Koa Shen

B.S. Candidate in Mechanical Engineering | UC Santa Barbara

koashensf@gmail.com (415) 810-8344 koashen.com San Francisco Bay Area / Santa Barbara, CA

October 2, 2026

Google Hardware Hiring Team

Re: Summer 2027 Hardware Engineering Internship

Dear Google Hiring Team,

I am a mechanical engineering student at UC Santa Barbara, graduating in June 2028, seeking a Summer 2027 hardware internship at Google. I am particularly interested in mechanical design, electromechanical integration, and hardware prototyping and validation. Google's devices appeal to me because useful technology depends on physical details that users should not have to think about: packaging, durable interfaces, consistent operation, and intuitive interaction. My experience combines simulation, precision fabrication, embedded controls, and maintaining hardware under continuous public use.

At SLAC National Accelerator Laboratory, I built a Python/Drake digital twin for a 30-DOF, 305-body beamline assembly. Millisecond collision checks identified at least five physical pinch points and guided mechanical redesigns. I worked on optics packaging, hot-swappable kinematic mounts, alignment features, and detector strain relief, incorporating instrument-scientist feedback and checking load cases around \$100,000-plus detectors. I also produced Solid Edge designs with GD&T and PDM-controlled documentation. The framework's 15-stage reusable catalog is now supporting adoption across three LCLS assemblies.

With UCSB Gaucho Racing, I led 15 teammates through driver safety-system design, manufacturing, and installation, with GR26 passing FSAE technical inspection. I CNC-machined 33 suspension components with critical bearing fits within 0.0005 inch and helped develop a steering wheel integrating controls, a display, and a quick-release mechanism. Separately, I designed a compact 15:1 cycloidal actuator and wrote C++ firmware for stepper control and encoder feedback, bench-validating position correction and missed-step detection on a single-joint prototype.

My Exploratorium work strengthened the user and reliability side of that foundation. I serviced more than 100 exhibits, completed three major overhauls, and redesigned parts that cracked, wore, or became difficult to maintain in daily use. I left CAD, CAM files, and remanufacturing instructions so future technicians could support those designs. At Google, I would bring that same attention to real operating conditions, along with the ability to move between CAD, code, fabrication, and testing while collaborating across disciplines.

Thank you for your consideration. I would welcome the opportunity to contribute to hardware that makes Google's technology dependable and useful in everyday life.

Sincerely,

Koa Shen